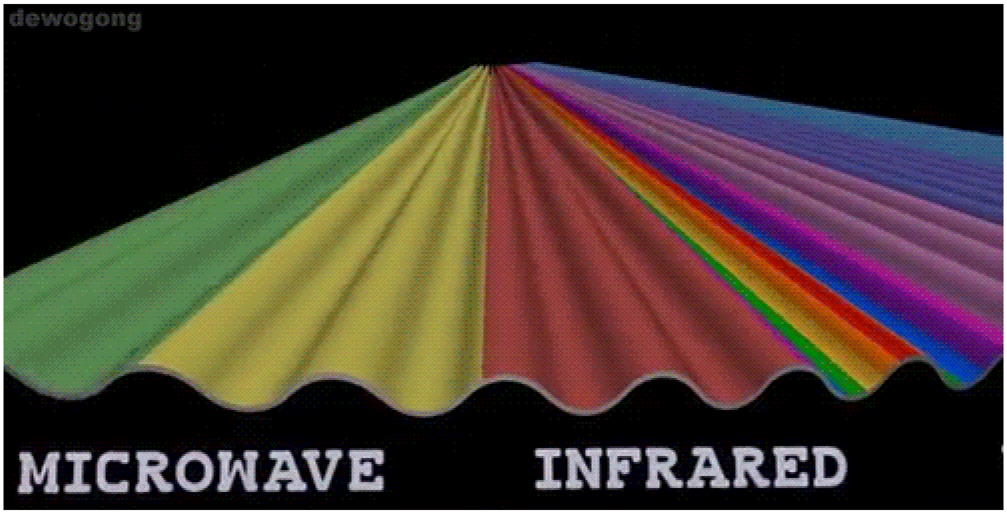
22_popquiz electromagnetic spectrum

This is a preview of the published version of the quiz

Started: Mar 23 at 6:43pm

Quiz Instructions



Question 1 10 pts

2) Electromagnetic waves consist of
A) oscillating electric and magnetic fields.
B) particles of light energy.
C) high-frequency sound waves.
D) compressions and rarefactions of electromagnetic pulses.

- C
- B
- A
- D

Question 2 10 pts

3) A changing electric field induces a changing
   A) magnetic field.
   B) electromagnetic field.
   C) both of these
   D) neither of these

○ B

○ D

○ A

○ C

Question 3 10 pts

4) In free space, electromagnetic waves travel at a
   A) speed depending on frequency.
   B) variety of speeds.
   C) single speed.

○ B

○ A

○ C

○ D

Question 4 10 pts

5) Which of the following is fundamentally different from the others?
   A) X–rays
   B) gamma rays
   C) sound waves
   D) radio waves
   E) light waves

○ D

- C
- A
- B

Question 5 10 pts

6) Which of these electromagnetic waves have the shortest wavelength?
A) X–rays
B) infrared waves
C) light waves
D) ultraviolet waves
E) radio waves

- A
- C
- B
- D

Question 6 10 pts

7) Which of these electromagnetic waves have the longest wavelength?
A) X–rays
B) radio waves
C) light waves
D) infrared waves
E) ultraviolet waves

- C
- A
- D

B

Question 7 10 pts

Consider a light source. Let's say an IR lamp used for reptiles or for therapy. The Power is 40W. The energy will flow in space in 3D. The same energy produced by the source will spread out over a sphere of radius R and area 4piR^2. The energy flowing through a unit area at a distance R is therefore P / (4piR^2)

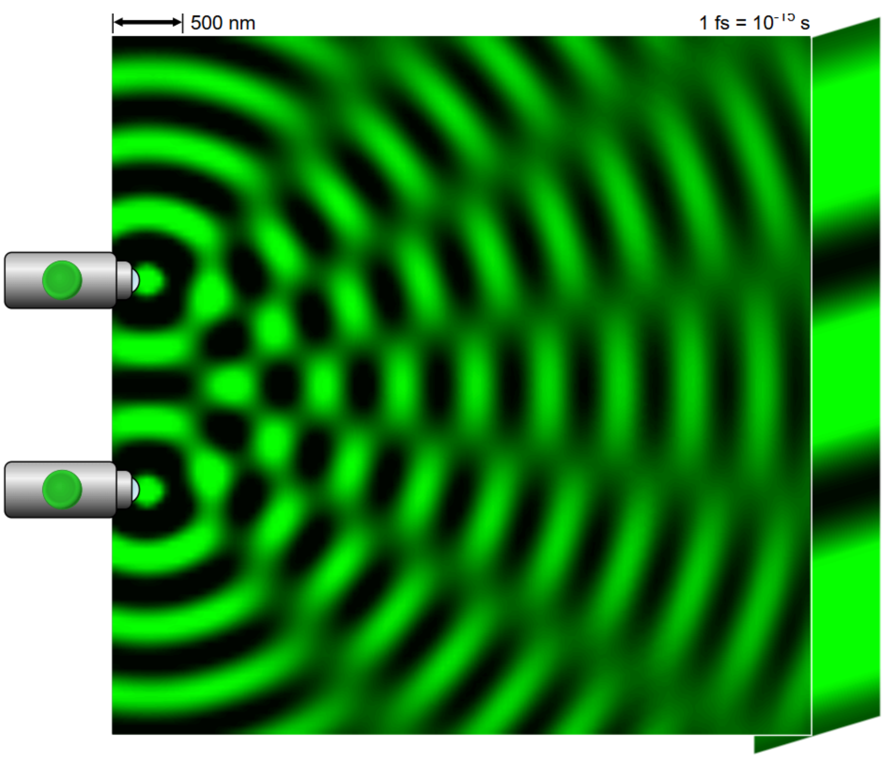
So at a distance R=50cm. The energy flowing through a unit area is:

(remember for final)

- about 13W/m^2
- about 8W/m^2
- about 40W /m^2
- bout 126W/m^2

Question 8 10 pts

That experiment (done with light.)shows which property of waves?



- ◯ interference
- ◯ polarization
- ◯ refraction
- ◯ reflection

Question 9 10 pts

$$\oint \vec{B} \cdot d\vec{a} = 0$$

This equation shows that

- ◯ the circulation of a static magnetic field is 0
- ◯ a magentic field always diverge or converge
- ◯ a magentic field circulates around a current
- ◯ there is no monopole

Question 10 10 pts

What do you find when you compute:

$$\frac{1}{\sqrt{\mu_0 \varepsilon_0}}$$

See constants in equation sheet.

- ◯ speed of light
- ◯ a random number

- [ ] the frequency of FM radio
- [ ] speed of light squared

Question 11 5 pts

$$\int \vec{E} \cdot d\vec{A} = 0$$ A

$$\int \vec{B} \cdot d\vec{A} = 0$$ C

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$ B

$$\oint \vec{B} \cdot dl = \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt}$$ D

Here are the very elegant Maxwell equations in space (no current)

Which one says that an magnetic field circulates around a changing electric field?

- [ ] D
- [ ] A
- [ ] C
- [ ] B

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